Week 1

Case Study 1

Data for Questions 1 to 5

Smart Classroom Ventilation Control

A university building uses a simple automatic ventilation controller for a smart classroom. The controller is intended to be transparent enough for maintenance staff to verify without running a learning algorithm. At the beginning of every class interval, the controller reads four binary indicators:

$$x_1 : \text{carbon dioxide level is high}, \quad x_2 : \text{room occupancy is high},$$

$$x_3 : \text{indoor temperature is above the comfort limit}, \quad x_4 : \text{outside air quality is acceptable}.$$

Each indicator is 1 when the condition is true and 0 otherwise. The ventilation fan is controlled by a McCulloch-Pitts style threshold unit:

$$y = \begin{cases} 1, & \text{if } x_1 + x_2 + x_3 + x_4 \geq \theta, \\ 0, & \text{otherwise.} \end{cases}$$

Here, $y = 1$ means that the fan is switched on. The facilities team wants the fan to switch on only when at least three of the four indicators are active. Each class interval is evaluated independently.

Question 1: Which threshold value implements the desired fan-control rule?

A) $\theta = 1$

B) $\theta = 2$

C) $\theta = 3$

D) $\theta = 4$

Correct Answer: C

Explanation: The fan should switch on when at least three indicators are active, so the threshold must be 3.

Question 2: For input $(1, 0, 1, 1)$, what output does the controller produce?

A) $y = 0$, because only one condition is absent

B) $y = 1$, because three indicators are active

C) $y = 0$, because all four indicators must be active

D) $y = 1$, because the first indicator is active

Correct Answer: B

Explanation: The input contains three 1s. Since $3 \geq \theta$, the output is $y = 1$.

Question 3: How many binary input patterns switch on the fan?

A) 5

B) 4

C) 6

D) 8

Correct Answer: A

Explanation: The fan switches on for exactly three active indicators or four active indicators. The number of patterns is $\binom{4}{3} + \binom{4}{4} = 4 + 1 = 5$.

Question 4: If the threshold is accidentally changed to $\theta = 2$, what happens?

A) The fan switches on less often than intended.

B) The fan switches on when two or more indicators are active.

C) The fan switches on only when all inputs are 1, making activation too strict.

D) The fan never switches on.

Correct Answer: B

Explanation: With $\theta = 2$, any pattern with two, three, or four active indicators produces $y = 1$, so activation becomes more frequent.

Question 5: Which property makes this controller a threshold-logic system?

A) It predicts future classroom occupancy states.

B) It updates weights after every incorrect output during a training process.

C) It requires a hidden layer to make a decision.

D) It compares a summed input value with a fixed threshold.

Correct Answer: D

Explanation: A threshold unit computes a weighted or unweighted sum and compares that sum with a threshold.

Case Study 2

Data for Questions 6 to 10

Scholarship Application Screening

A scholarship office builds a first-stage screening aid. The aid does not make the final award decision; it only flags applications that satisfy a simple rule for immediate review. Each application is represented by two binary features:

$$x_1 : \text{academic score is above the cutoff}, \quad x_2 : \text{financial-need certificate is verified.}$$

The office wants the system to flag an application when at least one of these two conditions is true. A perceptron with weights w_1, w_2, bias b, and decision rule

$$\hat{y} = \begin{cases} 1, & \text{if } w_1 x_1 + w_2 x_2 + b \geq 0, \\ 0, & \text{otherwise} \end{cases}$$

is tested for this screening rule.

Question 6: Which logical behavior is described by the desired screening rule?

A) AND

B) XOR

C) OR

D) XNOR

Correct Answer: C

Explanation: The output is 1 when $x_1 = 1$, or $x_2 = 1$, or both. This is the OR operation.

Question 7: Which parameter setting correctly implements this rule?

A) $w_1 = 1,\ w_2 = 1,\ b = -1.5$

B) $w_1 = 1,\ w_2 = 1,\ b = -0.5$

C) $w_1 = -1,\ w_2 = -1,\ b = 0.5$

D) $w_1 = 1,\ w_2 = -1,\ b = 0$

Correct Answer: B

Explanation: With $w_1 = w_2 = 1$ and $b = -0.5$, the score is negative only for $(0, 0)$. All other input patterns produce a non-negative score.

Question 8: For $w_1 = 1,\ w_2 = 1,\ b = -0.5$, what is the score for input $(0, 1)$?

A) 0.5

B) -0.5

C) 0

D) 1.5

Correct Answer: A

Explanation: The score is $1(0) + 1(1) - 0.5 = 0.5$, so the perceptron outputs 1.

Question 9: What is the decision boundary for the parameter setting $w_1 = 1,\ w_2 = 1,\ b = -0.5$?

A) $x_1 - x_2 - 0.5 = 0$

B) $x_1 + x_2 + 0.5 = 0$

C) $2x_1 + x_2 = 0$

D) $x_1 + x_2 - 0.5 = 0$

Correct Answer: D

Explanation: The boundary is where the perceptron score equals zero: $x_1 + x_2 - 0.5 = 0$.

Question 10: Why can a single perceptron represent this screening rule?

A) The positive and negative cases are linearly separable.

B) The rule requires remembering earlier applications before deciding the next output.

C) The rule has no decision boundary.

D) The output changes randomly for the same input.

Correct Answer: A

Explanation: OR is linearly separable because one straight line can separate $(0, 0)$ from the three positive input patterns.

Case Study 3

Data for Questions 11 to 15

Equipment Handover Record Review

An engineering lab tracks two binary checks before handing over a portable measurement kit:

x_1 : battery pack has been returned, x_2 : sensor module has been returned.

For audit purposes, the lab wants to flag a handover record when exactly one of the two items has been returned. If neither item has been returned or both items have been returned, the record is not flagged. The desired output table is:

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

The lab initially tries to implement this rule with a single perceptron.

Question 11: Which logical behavior does the handover rule represent?

A) AND

B) OR

C) XOR

D) NAND

Correct Answer: C

Explanation: The output is 1 exactly when the two binary inputs are different, which is the XOR rule.

Question 12: What output is assigned to the input pattern $(1, 1)$?

A) $y = 1$, because one item has been returned

B) $y = 0$, because exactly one item has not been returned

C) $y = 1$, because both checks are complete

D) $y = 0$, because both values are the same

Correct Answer: D

Explanation: The rule flags only records with exactly one returned item. For $(1, 1)$, both items have been returned, so $y = 0$.

Question 13: Which geometric property prevents a single perceptron from representing this rule?

A) The positive cases lie on opposite corners of the input square.

B) The inputs are binary rather than real valued, so the rule cannot use a separator.

C) The output has only two possible labels.

D) The rule uses two input variables.

Correct Answer: A

Explanation: The positive cases $(0, 1)$ and $(1, 0)$ are diagonally opposite, so a single straight line cannot separate them from the two negative cases.

Question 14: Which modified lab rule would be linearly separable?

A) Flag exactly one returned item.

B) Flag when the two item statuses are different.

C) Flag only when both items have been returned.

D) Flag when neither or both items have been returned.

Correct Answer: C

Explanation: Flagging only $(1, 1)$ is the AND rule, which is linearly separable.

Question 15: What does linear separability mean in this setting?

A) Every input must have label 1 before a straight boundary can be checked.

B) A line can separate the 1-labeled and 0-labeled inputs.

C) The perceptron must avoid using weights.

D) The input values must be stored in a sequence.

Correct Answer: B

Explanation: Linear separability means that a line can place all positive examples on one side and all negative examples on the other side.

Case Study 4

Data for Questions 16 to 20

Parcel Routing Decision Review

A courier service trains a perceptron to classify parcels into "express" and "standard" routing. Each parcel is described by two numerical features:

$$x_1 : \text{compactness score}, \quad x_2 : \text{priority-label score}.$$

The label is $y = 1$ for express routing and $y = -1$ for standard routing. The perceptron has no bias term and uses the update rule

$$w \leftarrow w + yx$$

whenever $y(w^\top x) \leq 0$. The training samples are processed in the following order:

Parcel	x_1	x_2	y
P_1	2	1	1
P_2	1	2	1
P_3	-1	-1	-1

Training starts with $w = (0, 0)$.

Question 16: What is the weight vector after processing P_1 once?

A) $w = (2, 1)$

B) $w = (1, 2)$

C) $w = (-1, -1)$

D) $w = (0, 0)$

Correct Answer: A

Explanation: Initially $w^\top x = 0$, so $y(w^\top x) = 0$ and an update is made. The new vector is $w = (0, 0) + 1(2, 1) = (2, 1)$.

Question 17: After the update from P_1, what is $w^\top x$ for P_2?

A) 2

B) 3

C) 4

D) 5

Correct Answer: C

Explanation: With $w = (2, 1)$ and $x = (1, 2)$, the score is $2(1) + 1(2) = 4$.

Question 18: With $w = (2, 1)$, is P_3 correctly classified?

A) No, because the raw score $w^\top x$ is greater than zero.

B) No, because $y(w^\top x) < 0$.

C) Yes, because $w^\top x = 0$.

D) Yes, because $y(w^\top x) > 0$.

Correct Answer: D

Explanation: For $P_3 = (-1, -1)$, $w^\top x = -3$. Since $y = -1$, $y(w^\top x) = 3 > 0$, so it is correctly classified.

Question 19: Which condition is needed for the perceptron convergence guarantee?

A) Every feature must be binary.

B) The data must be linearly separable.

C) The learning rate must be zero throughout all training updates.

D) All labels must be positive.

Correct Answer: B

Explanation: The perceptron convergence theorem applies when the training data is linearly separable.

Question 20: Which vector separates the three listed parcels?

A) $w = (-1, 1)$

B) $w = (1, -1)$

C) $w = (-1, -1)$

D) $w = (1, 1)$

Correct Answer: D

Explanation: With $w = (1, 1)$, P_1 and P_2 have positive scores, while P_3 has a negative score, matching the labels.